

CONSTANT *RM*

VARIABLES

rmState,
tmState,
tmPrepared,
msgs

Messages $\triangleq$

$[type : \{\text{"Prepared"}\}, rm : RM] \cup [type : \{\text{"Commit"}, \text{"Abort"}\}]$

TPTypeOK $\triangleq$

$\wedge rmState \in [RM \rightarrow \{\text{"working"}, \text{"prepared"}, \text{"committed"}, \text{"aborted"}\}]$
 $\wedge tmState \in \{\text{"init"}, \text{"done"}\}$
 $\wedge tmPrepared \subseteq RM$
 $\wedge msgs \subseteq Messages$

TPInit $\triangleq$

$\wedge rmState = [r \in RM \mapsto \text{"working"}]$
 $\wedge tmState = \text{"init"}$
 $\wedge tmPrepared = \{\}$
 $\wedge msgs = \{\}$

TMRcvPrepared(*r*) $\triangleq$

$\wedge tmState = \text{"init"}$
 $\wedge [type \mapsto \text{"Prepared"}, rm \mapsto r] \in msgs$
 $\wedge tmPrepared' = tmPrepared \cup \{r\}$
 $\wedge \text{UNCHANGED } \langle rmState, tmState, msgs \rangle$

TMCommit $\triangleq$

$\wedge tmState = \text{"init"}$
 $\wedge tmPrepared = RM$
 $\wedge tmState' = \text{"done"}$
 $\wedge msgs' = msgs \cup \{[type \mapsto \text{"Commit"}]\}$
 $\wedge \text{UNCHANGED } \langle rmState, tmPrepared \rangle$

TMAbort $\triangleq$

$\wedge tmState = \text{"init"}$
 $\wedge tmState' = \text{"done"}$
 $\wedge msgs' = msgs \cup \{[type \mapsto \text{"Abort"}]\}$
 $\wedge \text{UNCHANGED } \langle rmState, tmPrepared \rangle$

RMPprepare(*r*) $\triangleq$

$\wedge rmState[r] = \text{"working"}$
 $\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"prepared"}]$

$$\begin{aligned} &\wedge \text{msgs}' = \text{msgs} \cup \{[type \mapsto \text{"Prepared"}, rm \mapsto r]\} \\ &\wedge \text{UNCHANGED } \langle tmState, tmPrepared \rangle \end{aligned}$$

$$\begin{aligned} RMChooseToAbort(r) &\triangleq \\ &\wedge rmState[r] = \text{"working"} \\ &\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"aborted"}] \\ &\wedge \text{UNCHANGED } \langle tmState, tmPrepared, msgs \rangle \end{aligned}$$

$$\begin{aligned} RMRcvCommitMsg(r) &\triangleq \\ &\wedge [type \mapsto \text{"Commit"}] \in msgs \\ &\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"committed"}] \\ &\wedge \text{UNCHANGED } \langle tmState, tmPrepared, msgs \rangle \end{aligned}$$

$$\begin{aligned} RMRcvAbortMsg(r) &\triangleq \\ &\wedge [type \mapsto \text{"Abort"}] \in msgs \\ &\wedge rmState' = [rmState \text{ EXCEPT } ![r] = \text{"aborted"}] \\ &\wedge \text{UNCHANGED } \langle tmState, tmPrepared, msgs \rangle \end{aligned}$$

$$\begin{aligned} TPNext &\triangleq \\ &\vee TMCommit \vee TMAbort \\ &\vee \exists r \in RM : \\ &\quad TMRcvPrepared(r) \vee RMPprepare(r) \vee RMChooseToAbort(r) \\ &\quad \vee RMRcvCommitMsg(r) \vee RMRcvAbortMsg(r) \end{aligned}$$

$$TPSpec \triangleq TPInit \wedge \Box[TPNext]_{\langle rmState, tmState, tmPrepared, msgs \rangle}$$

THEOREM $TPSpec \Rightarrow \Box TPTypeOK$

INSTANCE $TCommit$

THEOREM $TPSpec \Rightarrow TCSpec$